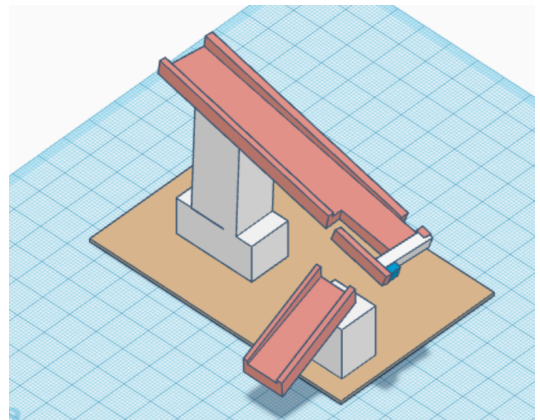
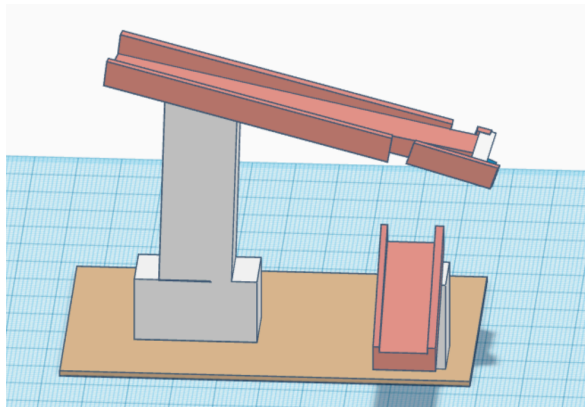


Problem Statement :-

Sorting objects by colour in industries can be a big job, and doing it by hand is slow and can lead to mistakes. That's why we're working on a project to create a simple and affordable Color Sorter. This machine can quickly and accurately sort things by their colour, making the sorting process much faster and more reliable for businesses. Our aim is to offer an easy-to-use solution that helps industries save time and avoid errors in sorting. This Color Sorter is designed to be practical and budget-friendly, providing a useful tool for businesses looking to improve their sorting operations

How the 3D model is designed:-



Base Creation:

We made the base of the model using the cube shape from the basic shapes menu. On the left side, we added support for the slide, creating a slant by using a triangle with a hole option.

Main Slide:

For the main slide, we used a cube shape.

We made the slide hollow by placing another cube with a hole option inside it.

Creating an opening at the bottom of the slide, we used a cube with a hole option. This is where the slide gate will open.

Slide Gate:

We designed the slide gate using a cube shape. This part controls the opening and closing of the slide.

Repeat for Smaller Slide:

We repeated the same process for the smaller slide. Using cubes and holes, we created a hollow structure with an opening for the smaller slide gate.

Servo Motors:

To represent the servo motors, we used small boxes with a blue colour. This helps visually denote the location of the motors in the model.

By using cubes and hole options creatively, we made this 3D model that includes the main base, two slides, slide gates, and servo motors. This design allows the colour sorter to function effectively in a simple and visually clear way.

STL file of the 3d model:-

<https://drive.google.com/drive/folders/1VCngbw0VubL4K2rfOwlAiEYNBgJdKbVo>

Tinkercad link:- <https://www.tinkercad.com/things/5daYTGf887q-color-sorter-3d-model>

Creation of the Actual model

Introduction:-

We made our colour sorter using simple cardboard for the structure. It's easy to work with, and we wanted to keep things straightforward.

To sort colours, we used balls in red, green, blue, and yellow. These colourful balls help our device recognize and sort different colours.

Cardboard might not be the strongest material, but it lets us quickly test and tweak our design. Plus, it's affordable and accessible for anyone wanting to try colour sorting.

Our goal is to make colour sorting fun and easy, even if you're just getting started with these kinds of projects.

Components Integration:

The 3D models helped us put together and secure different parts like the Arduino Nano, servo motors, jumper wires, veroboard, colour sensor/ detector, female power socket, male burg pin, 7803 regulator and a capacitor. They acted like a guide for everything to fit and work well.

Testing and Validation:

We checked that our design was strong and that all the parts fit well together. This ensured our colour sorter works the way it's supposed to.

Discussion on Circuit Diagram:-

We couldn't create a circuit diagram as the colour sensor/ detector is not available on tinkercad. But here is the explanation about the circuit

Power Supply:

The power supply provides power to the system. The regulator and capacitor ensure stable power.

Arduino Nano Connections:

D2 (S0), D3 (S1), D4 (S2), D5 (S3):

These pins are connected to the corresponding pins on the colour sensor (TCS230/TCS3200) to control the colour detection settings.

D6 (OUT): Connected to the OUT pin of the colour sensor to read the colour detection output frequency.

D7: Connected to the control input of the top servo motor (topServo) to simulate the gate opening and closing.

D8: Connected to the control input of the bottom servo motor (bottomServo) for adjusting its position based on the detected colour.

Top Servo Motor Connections:

Yellow Wire (D7): Connected to Arduino Nano D7 for control.

Red Wire (5V): Connected to the 5V power supply.

Black Wire (GND): Connected to the common ground.

Bottom Servo Motor Connections:

Yellow Wire (D8): Connected to Arduino Nano D8 for control.

Red Wire (5V): Connected to the 5V power supply.

Black Wire (GND): Connected to the common ground.

Color Sensor Connections:

VCC (5V): Connected to the 5V power supply.

GND: Connected to the common ground.

OUT (D6): Connected to Arduino Nano D6 for reading the colour detection output.

Explanation about components :-

1. SG90 Micro servo:- The SG90 micro servo is a small, lightweight motor that can be controlled precisely. In our colour sorter project, it acts as a bottom servo which is used to control the slide which helps the balls move in different directions.

2. MG90S mini digital servo motor:- Similar to the SG90, the MG90S is a mini digital servo motor. It is designed for small and lightweight applications, providing accurate and fast control of the rotational movement. In this project it is used for the opening and the closing of the cardboard for the ball to go onto the slide

3. TCS230/ TCS3200 Color recognition sensor/detector:- The TCS230/ TCS3200 colour sensor is a device that can detect and recognize colours. It consists of an array of photodiodes and colour filters. When exposed to light, the sensor can determine the intensity of red, green, and blue light, allowing it to identify the colour of an object. In our colour sorter, this sensor is crucial for identifying the colours of the balls mentioned (red, green, blue, and yellow).

4. Arduino Nano Board CH340/ATmega328P:- The Arduino Nano is a small, versatile microcontroller board based on the ATmega328P microcontroller. It is often used in projects where a compact size is essential. The Nano serves as the brain of our colour sorter, processing information from the colour sensor and controlling the servo motors based on the detected colours.

5. Jumper wires:- Jumper wires are electrical wires with connectors at each end, typically used to interconnect components on a breadboard or circuit. In our project, jumper wires facilitate the connection between the Arduino Nano, servo motors, and the colour sensor, establishing the necessary electrical pathways for data and power.

6. Female Power Socket :- A female power socket is a connector that allows you to easily connect and disconnect a power source. In our project, it likely serves as the point where we can plug in an external power supply to provide power to the circuit.

7. Male Berg pin:- A male Berg pin (commonly referred to as a male header or pin) is a component with protruding pins that can be inserted into a corresponding female connector. In electronics, it is often used to establish secure connections on a circuit board. In our project we used it for the connection with Arduino Nano and to connect the servo motor and colour sensor.

8. 7803 regulator :- The 7803 regulator is a type of voltage regulator. Specifically, it is a positive voltage regulator that provides a stable and fixed output voltage of 3 volts. In our circuit, it plays a role in regulating the voltage supplied to specific components, ensuring they receive a consistent and appropriate voltage level.

9. Capacitor:- A capacitor is an electronic component that stores electrical energy in an electric field. In our project, it is used for smoothing out voltage fluctuations or filtering unwanted signals.

10. Veroboard:- Veroboard (also known as stripboard) is a type of circuit board with pre-arranged copper tracks. It is used for creating prototypes or permanent circuit designs by soldering components onto the board. In our project, the Veroboard serves as the platform for mounting and connecting the various components, providing a structured layout for our circuit.

Explanation of the code:-

Libraries and Definitions:-

```
#include <Servo.h>
#define S0 2
#define S1 3
#define S2 4
#define S3 5
#define sensorOut 6
Servo topServo;
Servo bottomServo;
int frequency = 0;
int colour=0;
boolean GATE_FLG=0;
```

- The code starts by including the Servo library for controlling servo motors.
- It defines pin numbers for the colour sensor (S0-S3), sensor output (sensorOut), and creates instances of two servo motors (topServo and bottomServo).
- Frequency and colour are variables to store the frequency of colour detection and the identified colour, respectively. GATE_FLG is a boolean flag indicating whether the sorting gate is open.

Setup Function:-

```
void setup() { //01
  pinMode(S0, OUTPUT);
  pinMode(S1, OUTPUT);
  pinMode(S2, OUTPUT);
  pinMode(S3, OUTPUT);
  pinMode(sensorOut, INPUT);
  // Setting frequency-scaling to 20%
```

```
digitalWrite(S0, HIGH);  
digitalWrite(S1, LOW);  
topServo.attach(7);  
bottomServo.attach(8);  
Serial.begin(9600);  
}
```

- The setup() function initialises pin modes for sensor control pins and sensor output.
- It sets the frequency scaling of the colour sensor (S0 and S1) to 20% (HIGH and LOW).
- Attaches the servo motors to pins 7 and 8.
- Initialises serial communication for debugging.

Loop Function:

```
void loop()  
  
colour = readColor();  
  
delay(10);  
  
switch(colour){  
  
    case 1:  
  
        bottomServo.write(30); GATE_FLG=1;  
  
        break;  
  
    case 2:  
  
        bottomServo.write(70); GATE_FLG=1;  
  
        break;  
  
    case 3:  
  
        bottomServo.write(115);GATE_FLG=1;  
  
        break;  
  
    case 4:
```

```

    bottomServo.write(160);GATE_FLG=1;

    break;

    case 0:

    break;

}

colour=0;

if(GATE_FLG){

    delay(200);

    topServo.write(0);

    delay(280);

    topServo.write(90);

    delay(1000);

    GATE_FLG=0;

}

```

- The loop() function reads the colour using the readColor() function.
- Based on the detected colour, it adjusts the position of the bottomServo to sort the object.
- It also controls the topServo to simulate a gate opening and closing, providing a delay before resetting the gate flag.

Custom Function - readColor():

```

int readColor() {

    digitalWrite(S2, LOW);

    digitalWrite(S3, LOW);

```



```
frequency = pulseIn(sensorOut, LOW);

int R = frequency;

Serial.print("R= ");

Serial.print(frequency); Serial.print(" ");

delay(50);

digitalWrite(S2, HIGH);

digitalWrite(S3, HIGH);

frequency = pulseIn(sensorOut, LOW);

int G = frequency;

Serial.print("G= ");

Serial.print(frequency);

Serial.print(" ");

delay(50);

digitalWrite(S2, LOW);

digitalWrite(S3, HIGH);

frequency = pulseIn(sensorOut, LOW);

int B = frequency;

Serial.print("B= ");

Serial.print(frequency);
```

```
Serial.println(" ");

delay(50);

if((R>16 & R<19) & (G>55 & G<58) & (B>52 & B<55)){

    colour = 1;    // Red

    Serial.println(" RED ");

}

if(R>31 & R<45 & G>38 & G<40 & B>51 & B<58){

    colour = 2;    // Green

    Serial.println(" GREEN ");

}

if(R>12 & R<14 & G>20 & G<21 & B>34 & B<36){

    colour = 3; Serial.println(" YELLOW "); // Yellow

}

if(R>40 & R<43 & G>50 & G<54 & B>38 & B<42){

    colour = 4; // Blue

    Serial.println(" BLUE ");

}

return colour;

}
```

- The readColor() function configures the colour sensor for red, green, and blue readings.
- It captures the frequency values for each colour channel (R, G, B) and prints them to the serial monitor.
- Based on the frequency values, it identifies the colour (red, green, yellow, blue) and returns the corresponding colour code.

The overall functionality involves colour detection using the TCS230/TCS3200 sensor, and the sorting mechanism is controlled by servo motors based on the identified colour. The gate mechanism is simulated with the topServo, and the sorting action is performed by adjusting the position of the bottomServo. The colour identification is based on predefined frequency ranges for each colour. The serial monitor is used for debugging and displaying the colour identification process.

REFERENCES

“How to Make Arduino Colour Sorting Machine - Cadbury Gems - Indian Lifehacker.” 2018. *YouTube*. YouTube. July 1. <https://www.youtube.com/watch?v=4DbrWAGDAD>

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